1. What mathematical operator represents the spatial rate of increase of a scalar field?
- **(A)** Divergence (div)
- **(B)** Gradient (grad)
- **(C)** Curl
- **(D)** Laplacian

2. The divergence of a vector field Del dot F physically represents:
- **(A)** Vorticity around a point
- **(B)** Maximum directional slope
- **(C)** Vector cross product
- **(D)** Net outward flux per unit volume

3. A vector field whose divergence is zero everywhere is termed:
- **(A)** Solenoidal field
- **(B)** Irrotational field
- **(C)** Conservative field
- **(D)** Laminar field

4. A vector field whose curl is identically zero everywhere is known as:
- **(A)** Solenoidal field
- **(B)** Rotational field
- **(C)** Irrotational field
- **(D)** Divergent field

5. Which theorem relates a volume integral of divergence to a closed surface integral?
- **(A)** Gauss's Divergence Theorem
- **(B)** Stokes's Theorem
- **(C)** Green's Theorem
- **(D)** Ampere's Circuital Law

6. Stokes's theorem transforms a surface integral of curl into a:
- **(A)** Volume integral of flux
- **(B)** Surface integral of gradient
- **(C)** Closed line integral (circulation)
- **(D)** Point potential function

7. The continuity equation in electrodynamics is a direct expression of:
- **(A)** Conservation of momentum
- **(B)** Conservation of electric charge
- **(C)** Conservation of energy
- **(D)** Faraday's induction law

8. The differential form of the continuity equation relates Del dot J to:
- **(A)** + d(rho) / dt
- **(B)** 0
- **(C)** - Del^2(rho)
- **(D)** - d(rho) / dt

9. In steady DC circuits where charge density rho does not change with time, Del dot J equals:
- **(A)** mu_0 * J
- **(B)** epsilon_0
- **(C)** Zero
- **(D)** Infinity

10. Why was Ampere's original circuital law inconsistent for time-varying fields?
- **(A)** Divergence of curl is zero, but div J is not zero
- **(B)** It violated Coulomb's law
- **(C)** Magnetic fields have sources
- **(D)** Electric fields cannot curl

11. Who introduced the displacement current to resolve Ampere's circuital inconsistency?
- **(A)** Heinrich Hertz
- **(B)** Michael Faraday
- **(C)** Carl Friedrich Gauss
- **(D)** James Clerk Maxwell

12. Displacement current density J_d is mathematically defined as:
- **(A)** dE / dt
- **(B)** dD / dt
- **(C)** dB / dt
- **(D)** mu_0 * dH / dt

13. In free space, displacement current density J_d equals:
- **(A)** mu_0 * (dH / dt)
- **(B)** epsilon_0 * (dE / dt)
- **(C)** sigma * E
- **(D)** Del x B

14. Unlike conduction current, displacement current produces a magnetic field without:
- **(A)** Physical drift of charged particles
- **(B)** Time-varying electric fields
- **(C)** Permittivity of medium
- **(D)** Any associated energy flow

15. Maxwell's first equation (Gauss's Law for electrostatics) in differential form is:

(A) Del x E = - dB / dt
(B) Del dot B = 0
(C) Del dot D = rho_free
(D) Del x H = J

16. Maxwell's second equation Del dot B = 0 physically signifies:

(A) Magnetic fields are conservative
(B) Electric charge is quantized
(C) Displacement current is zero
(D) Absence of isolated magnetic monopoles

17. Maxwell's third equation (Faraday's Law) in differential form is:

(A) Del x E = 0
(B) Del dot E = rho / epsilon
(C) Del x B = mu * J
(D) Del x E = - dB / dt

18. Maxwell's fourth equation (Ampere-Maxwell Law) in differential form is:

(A) Del x H = J_free
(B) Del dot H = 0
(C) Del x H = J_free + dD / dt
(D) Del x E = - dD / dt

19. The SI unit of electric displacement vector D is:

(A) Coulombs per square meter (C/m^2)
(B) Volts per meter (V/m)
(C) Amperes per meter (A/m)
(D) Webers per square meter (Wb/m^2)

20. The SI unit of magnetic field intensity H is:

(A) Tesla (T)
(B) Amperes per meter (A/m)
(C) Webers (Wb)
(D) Henries per meter (H/m)

21. The relation connecting electric displacement D and electric field E in linear media is:

(A) D = E / epsilon
(B) D = epsilon * E
(C) D = mu * E
(D) D = sigma * E

22. The relation connecting magnetic flux density B and magnetic field H is:

(A) B = H / mu
(B) B = epsilon * H
(C) B = Del x H
(D) B = mu * H

23. In a charge-free, non-conducting dielectric medium (rho=0, J=0), Del dot E equals:

(A) Zero
(B) rho / epsilon_0
(C) dD / dt
(D) Infinity

24. The wave equation for electric field E in a vacuum is:

(A) Del(E) = 0
(B) Del^2(E) = - dE / dt
(C) Del^2(E) = mu_0 * epsilon_0 * (d^2E / dt^2)
(D) Del x E = mu_0 * epsilon_0 * E

25. The speed of electromagnetic waves in a vacuum is given by:

(A) c = 1 / sqrt(mu_0 * epsilon_0)
(B) c = sqrt(mu_0 / epsilon_0)
(C) c = mu_0 * epsilon_0
(D) c = sqrt(mu_0 * epsilon_0)

26. The numerical value of the speed of light c in vacuum is approximately:

(A) 3.0 x 10^6 m/s
(B) 1.5 x 10^8 m/s
(C) 3.0 x 10^8 m/s
(D) 3.0 x 10^10 m/s

27. Electromagnetic waves are fundamentally classified as:

(A) Longitudinal waves
(B) Transverse waves
(C) Torsional waves
(D) Mechanical pressure waves

28. In a plane EM wave propagating in free space along the +z axis, E and H are:

(A) Parallel to each other
(B) Along the z axis
(C) Opposite to each other
(D) Perpendicular to each other and to the z axis

29. The ratio of electric field magnitude E to magnetic field magnitude H in free space is:
- **(A)** Permittivity (epsilon_0)
- **(B)** Permeability (mu_0)
- **(C)** Intrinsic impedance (eta_0)
- **(D)** Refractive index (n)

30. The value of intrinsic impedance of free space eta_0 is approximately:
- **(A)** 377 Ohms
- **(B)** 120 Ohms
- **(C)** 50 Ohms
- **(D)** 600 Ohms

31. The Poynting vector S represents the directional rate of:
- **(A)** Electric charge flow per unit time
- **(B)** Magnetic flux linkage per turn
- **(C)** Mechanical force per unit volume
- **(D)** Electromagnetic energy flow per unit area

32. The mathematical expression for the instantaneous Poynting vector S is:
- **(A)** S = E dot H
- **(B)** S = E x H
- **(C)** S = B x H
- **(D)** S = E x D

33. The SI unit of the Poynting vector S is:
- **(A)** Joules per cubic meter (J/m^3)
- **(B)** Watts per square meter (W/m^2)
- **(C)** Volts per meter (V/m)
- **(D)** Newtons per Coulomb (N/C)

34. The time-average Poynting vector S_avg for a sinusoidal plane wave in free space is:
- **(A)** 0.5 * (E_0^2 / eta_0)
- **(B)** E_0^2 * eta_0
- **(C)** E_0 * H_0
- **(D)** 2 * E_0^2 / eta_0

35. Electromagnetic energy density stored in an electrostatic field E is given by:
- **(A)** epsilon * E
- **(B)** 0.5 * epsilon^2 * E
- **(C)** 0.5 * epsilon * E^2
- **(D)** 2 * epsilon * E^2

36. Electromagnetic energy density stored in a magnetostatic field B is given by:
- **(A)** mu * B^2
- **(B)** 0.5 * mu * B^2
- **(C)** B^2 / (2 * mu^2)
- **(D)** 0.5 * B^2 / mu

37. In a plane EM wave in vacuum, the average electric energy density is:
- **(A)** Twice the magnetic energy density
- **(B)** Half the magnetic energy density
- **(C)** Zero
- **(D)** Equal to the average magnetic energy density

38. The depth in a conducting medium where wave amplitude drops to 1/e of its surface value is:
- **(A)** Coherence length
- **(B)** Extinction depth
- **(C)** Skin depth (delta)
- **(D)** Debye length

39. In a good conductor, skin depth delta varies with frequency f as:
- **(A)** 1 / sqrt(f)
- **(B)** sqrt(f)
- **(C)** 1 / f
- **(D)** f^2

40. Ground Penetrating Radar (GPR) used in civil structural inspections relies on:
- **(A)** Nuclear acoustic resonance
- **(B)** Reflection of EM pulses at dielectric interfaces
- **(C)** Piezoelectric ultrasound radiation
- **(D)** Static magnetic induction

41. If a scalar potential is V = x^2 + 2y^2 + 3z^2, what is the electric field E at (1, 1, 1)?
- **(A)** +2i + 4j + 6k
- **(B)** -2i - 4j - 6k
- **(C)** -i - 2j - 3k
- **(D)** 0

42. For a vector field F = x^2 i + y^2 j + z^2 k, what is the divergence Del dot F at (2, 3, 4)?
- **(A)** 9
- **(B)** 24
- **(C)** 12
- **(D)** 18

43. For a field F = (2xy) i + (x^2 + 2yz) j + (y^2) k, the curl Del x F is:

(A) 0 (conservative field)
(B) 2x i + 2y j
(C) 4xy k
(D) -2z i

44. A parallel plate capacitor with circular plates of radius 0.1 m has dE/dt = 10^12 V/(m*s). The displacement current is:

(A) 0.028 A
(B) 2.78 A
(C) 0.278 A
(D) 27.8 A

45. A charging capacitor has conduction current i_c = 1.5 A entering the plates. The total displacement current between plates is:

(A) 1.5 A
(B) 0 A
(C) 0.75 A
(D) 3.0 A

46. In a non-magnetic lossless medium with relative permittivity epsilon_r = 4, the phase velocity of EM waves is:

(A) 3.0 x 10^8 m/s
(B) 0.75 x 10^8 m/s
(C) 1.5 x 10^8 m/s
(D) 6.0 x 10^8 m/s

47. The refractive index n of a non-magnetic medium with relative permittivity epsilon_r = 9 is:

(A) 9.0
(B) 3.0
(C) 4.5
(D) 1.73

48. The intrinsic impedance of a non-magnetic medium with epsilon_r = 4 is:

(A) 377 Ohms
(B) 754 Ohms
(C) 94.2 Ohms
(D) 188.5 Ohms

49. A plane EM wave in free space has electric field amplitude E_0 = 100 V/m. The magnetic field amplitude H_0 is:

(A) 26.5 A/m
(B) 37.7 A/m
(C) 0.265 A/m
(D) 0.0265 A/m

50. For the EM wave in Q49, the amplitude of magnetic flux density B_0 is:

(A) 3.33 x 10^-7 T
(B) 1.0 x 10^-4 T
(C) 3.0 x 10^10 T
(D) 8.85 x 10^-12 T

51. A plane wave has peak electric field E_0 = 60 V/m in free space. The time-average Poynting vector magnitude is:

(A) 9.55 W/m^2
(B) 2.38 W/m^2
(C) 19.1 W/m^2
(D) 4.77 W/m^2

52. Total electromagnetic energy density for a wave with E_rms = 50 V/m in air is:

(A) 4.42 x 10^-8 J/m^3
(B) 2.21 x 10^-8 J/m^3
(C) 1.10 x 10^-8 J/m^3
(D) 8.85 x 10^-8 J/m^3

53. For copper with conductivity sigma = 5.8 x 10^7 S/m and mu_r = 1, the skin depth at 50 Hz is approximately:

(A) 0.66 mm
(B) 9.35 mm
(C) 1.25 mm
(D) 66.0 mm

54. For copper at high frequency f = 1 GHz, the skin depth delta reduces to approximately:

(A) 2.09 micrometers
(B) 20.9 micrometers
(C) 0.21 micrometers
(D) 2.09 mm

55. In moist concrete (sigma = 0.05 S/m, epsilon_r = 9), the loss tangent tan(delta) at 100 MHz is approximately:

(A) 0.10
(B) 10.0
(C) 1.00
(D) 0.01

56. The continuity equation applied to a steady current state (d(rho)/dt = 0) reduces to:

(A) Del x J = 0
(B) Del^2 J = 0
(C) J = constant
(D) Del dot J = 0

57. If the magnetic field of an EM wave is H = 0.2*cos(omega*t - beta*z) i A/m, the electric field E is directed along:

(A) +j direction
(B) +k direction
(C) -i direction
(D) -j direction

58. What is the phase constant beta of a 100 MHz EM wave propagating in free space?

(A) 1.05 rad/m
(B) 3.14 rad/m
(C) 2.09 rad/m
(D) 6.28 rad/m

59. What is the wavelength lambda of a 1.5 GHz GPR radar signal in air?

(A) 0.20 m
(B) 0.10 m
(C) 0.50 m
(D) 1.50 m

60. When the 1.5 GHz GPR signal enters concrete with relative permittivity epsilon_r = 6.25, its wavelength becomes:

(A) 0.20 m
(B) 0.08 m
(C) 0.50 m
(D) 0.04 m

61. The propagation velocity of a GPR wave in saturated soil with epsilon_r = 25 is:

(A) 1.50 x 10^8 m/s
(B) 0.60 x 10^8 m/s
(C) 3.00 x 10^8 m/s
(D) 0.12 x 10^8 m/s

62. The divergence of a magnetic field B = (3x) i + (2y) j - (c*z) k must be zero. The value of constant c is:

(A) 1
(B) -5
(C) 0
(D) 5

63. The integral form of Faraday's Law states that the line integral of E around a closed loop equals:

(A) - d(Phi_B) / dt
(B) + d(Phi_E) / dt
(C) mu_0 * I
(D) Q / epsilon_0

64. Poynting's theorem can be viewed as an electromagnetic statement of:

(A) Momentum conservation
(B) Gauss's law
(C) Work-energy theorem for fields
(D) Lorentz force law

65. In a good conductor, the phase angle between electric field E and magnetic field H is:

(A) 45 degrees (pi/4 rad)
(B) 0 degrees
(C) 90 degrees (pi/2 rad)
(D) 180 degrees

66. The attenuation constant alpha in a good conductor is equal to:

(A) 2 * beta
(B) beta / 2
(C) beta = sqrt(pi * f * mu * sigma)
(D) zero

67. An EM wave travels through a dielectric with mu_r = 1 and epsilon_r = 2.25. The wave velocity is:

(A) 1.5 x 10^8 m/s
(B) 2.0 x 10^8 m/s
(C) 3.0 x 10^8 m/s
(D) 2.5 x 10^8 m/s

68. A laser beam carries an intensity of 1.0 kW/m^2. The peak electric field amplitude E_0 is approximately:

(A) 614 V/m
(B) 1228 V/m
(C) 275 V/m
(D) 868 V/m

69. For the laser beam in Q68, the peak magnetic field intensity H_0 is:

(A) 1.63 A/m
(B) 4.60 A/m
(C) 2.30 A/m
(D) 0.82 A/m

70. The radiation pressure exerted by a plane EM wave on a completely absorbing surface at normal incidence is:

(A) S_avg / c
(B) 2 * S_avg / c
(C) S_avg / (2*c)
(D) S_avg * c

71. If the surface is completely reflecting, the radiation pressure at normal incidence is:

(A) S_avg / c
(B) 4 * S_avg / c
(C) 0
(D) 2 * S_avg / c

72. A radio station broadcasts 50 kW isotropically. The average Poynting vector magnitude at distance 5 km is:

(A) 3.18 x 10^-4 W/m^2
(B) 1.59 x 10^-4 W/m^2
(C) 0.79 x 10^-4 W/m^2
(D) 6.36 x 10^-4 W/m^2

73. The electric field of a wave is E(z,t) = 150*cos(10^8 t - beta*z) i V/m. The frequency f of the wave is:

(A) 100 MHz
(B) 15.9 MHz
(C) 31.8 MHz
(D) 50 MHz

74. In seawater (sigma = 4 S/m, mu_r = 1), what is the skin depth at submarine communications frequency f = 100 Hz?

(A) 25.2 m
(B) 2.52 m
(C) 252 m
(D) 0.25 m

75. At microwave frequency f = 10 GHz in seawater, the skin depth delta decreases dramatically to:

(A) 25.2 mm
(B) 0.25 mm
(C) 2.52 mm
(D) 25.2 cm

76. A GPR antenna emits a pulse at a concrete bridge deck. The two-way travel time to a reinforcing bar is 2.4 ns. If epsilon_r = 9, the depth of the rebar is:

(A) 24.0 cm
(B) 6.0 cm
(C) 18.0 cm
(D) 12.0 cm

77. A void beneath a 30 cm thick concrete slab reflects a GPR signal with two-way time 5.0 ns. The effective dielectric constant of the slab is:

(A) 4.00
(B) 9.00
(C) 2.50
(D) 6.25

78. In structural GPR surveying, the reflection coefficient Gamma at an interface from concrete (epsilon_r1 = 9) to air void (epsilon_r2 = 1) is:

(A) -0.50 (reversed)
(B) +0.25
(C) +0.50 (in-phase)
(D) -0.75

79. At an interface between concrete (epsilon_r1 = 9) and steel rebar (treated as perfect conductor Gamma = -1.0), the reflected wave has:

(A) Complete reflection with 180 degree phase reversal
(B) Complete transmission without reflection
(C) Zero reflection coefficient
(D) In-phase enhancement

80. A parallel-plate capacitor has area A = 0.05 m^2 and plate separation d = 2 mm. It is driven by V(t) = 311*sin(314*t) V. The peak displacement current is:

(A) 3.90 x 10^-5 A
(B) 1.95 x 10^-5 A
(C) 0.98 x 10^-5 A
(D) 6.28 x 10^-5 A

81. For the capacitor in Q80, what is the peak magnetic field B induced between the plates at radius r = 5 cm?

(A) 3.90 x 10^-11 T
(B) 7.80 x 10^-11 T
(C) 1.56 x 10^-10 T
(D) 0

82. A plane wave in a lossy soil has attenuation constant alpha = 0.2 Np/m. How far can the wave penetrate before its power drops to 1% (-20 dB)?

(A) 5.0 m
(B) 23.0 m
(C) 2.3 m
(D) 11.5 m

83. A high-voltage power transmission line creates an electric field E = 5.0 kV/m at ground level (50 Hz). The displacement current density in air is:

(A) 1.39 x 10^-4 A/m^2
(B) 2.78 x 10^-4 A/m^2
(C) 0.70 x 10^-4 A/m^2
(D) 4.42 x 10^-4 A/m^2

84. If the soil underneath the transmission line in Q83 has conductivity sigma = 0.01 S/m, the conduction current density J_c in the soil is:

(A) 5.0 A/m^2
(B) 0.50 A/m^2
(C) 50.0 A/m^2
(D) 500 A/m^2

85. A 500 MHz GPR wave in saturated sandy soil (epsilon_r = 16, sigma = 0.02 S/m) has attenuation constant alpha equal to:

(A) 0.942 Np/m
(B) 1.884 Np/m
(C) 0.471 Np/m
(D) 3.14 Np/m

86. For the wave in Q85, the phase constant beta is:

(A) 20.9 rad/m
(B) 83.8 rad/m
(C) 41.9 rad/m
(D) 10.5 rad/m

87. What is the penetration depth (1/alpha) of the 500 MHz GPR signal in this saturated sand?

(A) 0.53 m
(B) 1.06 m
(C) 2.12 m
(D) 4.24 m

88. A coaxial cable has inner conductor radius a = 1 mm and outer conductor radius b = 5 mm with Teflon (epsilon_r = 2.1). Its characteristic impedance Z_0 is:

(A) 50.0 Ohms
(B) 75.0 Ohms
(C) 100.0 Ohms
(D) 66.5 Ohms

89. A solar panel on a bridge monitoring station receives solar irradiance S_avg = 900 W/m^2. The root-mean-square electric field E_rms of sunlight is:

(A) 824 V/m
(B) 412 V/m
(C) 582 V/m
(D) 291 V/m

90. For the sunlight in Q89, the radiation force on a flat 2.0 m^2 perfectly absorbing solar collector is:

(A) 6.0 x 10^-6 N
(B) 1.2 x 10^-5 N
(C) 3.0 x 10^-6 N
(D) 1.8 x 10^-5 N

91. In an electromagnetic shielding room for structural data acquisition, a 1 mm aluminum plate (sigma = 3.5 x 10^7 S/m) is used. At 1 MHz, the number of skin depths in the plate is:

(A) 6.0 skin depths
(B) 24.0 skin depths
(C) 3.0 skin depths
(D) 12.0 skin depths

92. The electromagnetic wave attenuation through the 1 mm aluminum shield in Q91 is approximately:

(A) 52 dB
(B) 104 dB
(C) 208 dB
(D) 26 dB

93. A GPS satellite transmits at L1 frequency 1575.42 MHz. The phase delay through an ionospheric layer with electron density N_e = 10^17 electrons/m^2 causes an apparent ranging error of approximately:

(A) 3.24 m
(B) 1.62 m
(C) 0.81 m
(D) 6.48 m

94. A radar antenna radiates 10 kW peak power uniformly over a hemisphere. What is the power density at a target bridge pier 2 km away?

(A) 3.98 x 10^-4 W/m^2
(B) 1.99 x 10^-4 W/m^2
(C) 7.96 x 10^-4 W/m^2
(D) 0.99 x 10^-4 W/m^2

95. If the radar target in Q94 has radar cross section sigma_rcs = 15 m^2, the power reradiated by the target is:

(A) 2.98 x 10^-3 W
(B) 1.19 x 10^-2 W
(C) 5.97 x 10^-3 W
(D) 5.97 x 10^-4 W

96. A 100 MHz plane wave travels from dry concrete (epsilon_r1 = 4.0) into saturated soil (epsilon_r2 = 16.0) at normal incidence. The transmission coefficient tau for electric field is:

(A) 0.33
(B) 1.33
(C) 0.50
(D) 0.67

97. For the interface in Q96, the percentage of incident power reflected back into the dry concrete is:

(A) 25.0%
(B) 33.3%
(C) 5.5%
(D) 11.1%

98. A cylindrical conductor carries total AC current I_0*cos(omega*t). Due to skin effect, the current density J(r) at distance r from the surface is governed by:

(A) J(r) = J_0 * (1 - r/delta)
(B) J(r) = J_0 / (1 + r/delta)
(C) J(r) = J_0 * exp(-r/delta) * cos(omega*t - r/delta)
(D) J(r) = J_0 * cos(omega*t)

99. In an underground pipeline cathodic protection system, an alternating surge creates a magnetic field H = 10 A/m in surrounding clay (sigma = 0.1 S/m, mu_r = 1) at 50 Hz. The induced electric field E at the pipe surface is:

(A) 0.199 V/m
(B) 1.99 V/m
(C) 0.020 V/m
(D) 0.995 V/m

100. The complex Poynting vector S_c = 0.5 * (E x H*) inside a lossy structural material has a non-zero imaginary part, which physically represents:

(A) Dissipated ohmic heat per cycle
(B) Reactive or stored oscillating energy density
(C) Static mechanical stress tensor
(D) Charge accumulation rate

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Define scalar and vector fields, gradient, divergence, and curl. Give their mathematical expressions in Cartesian coordinates and state their physical significance in electromagnetism.

Q2. State and prove Gauss's Divergence Theorem. Explain its role in transitioning from Maxwell's integral to differential equations.

Q3. State and prove Stokes's Theorem. Demonstrate how it connects Faraday's Law in integral and differential forms.

Q4. Derive the Equation of Continuity from the fundamental principle of charge conservation. Explain its physical significance in time-varying fields.

Q5. Explain the inconsistency of Ampere's Circuital Law for time-dependent fields. How did Maxwell resolve this dilemma by introducing the displacement current?

Q6. Describe the parallel-plate capacitor paradox during charging, and prove mathematically that conduction current in the leads equals displacement current between the plates.

Q7. State all four Maxwell's equations in both differential and integral forms for general media, and explain the physical law underlying each equation.

Q8. Derive the electromagnetic wave equation for electric field E in a linear, isotropic, homogeneous, and charge-free dielectric medium.

Q9. Derive the electromagnetic wave equation for magnetic field H in a charge-free dielectric, and compare its phase velocity with that of the electric field.

Q10. Prove that plane electromagnetic waves in free space are strictly transverse in nature (i.e., both E and H have zero components along the direction of propagation).

Q11. Derive the relationship between electric and magnetic field amplitudes in a plane EM wave, and obtain the expression for the intrinsic impedance of free space.

Q12. State and derive Poynting's Theorem from Maxwell's equations. Provide the physical interpretation of each term in the resulting equation.

Q13. Calculate the time-average Poynting vector for a harmonic plane electromagnetic wave, and prove that energy is equally partitioned between electric and magnetic fields.

Q14. Derive the electromagnetic wave equation in a conducting medium, and obtain expressions for the attenuation constant alpha and phase constant beta.

Q15. Define skin depth (depth of penetration). Derive its formula for a good conductor and explain its engineering consequences in structural grounding and EM shielding.

Q16. Explain the working principles of Ground Penetrating Radar (GPR) in civil engineering. How do permittivity and conductivity govern radar depth and resolution?

Q17. A circular parallel-plate capacitor of radius R = 15 cm and plate separation d = 3.0 mm is charged by an AC source V(t) = 240*sqrt(2)*sin(100*pi*t) V. Determine: (a) capacitance C, (b) conduction current i_c(t), (c) displacement current density J_d(t), (d) induced magnetic field B at radius r

Q18. A plane electromagnetic wave in air has electric field E(z,t) = 120*cos(6*pi*10^8 t - beta*z) i V/m. Calculate: (a) frequency f and wavelength lambda, (b) phase constant beta, (c) magnetic field expression H(z,t), (d) time-average Poynting vector S_avg.

Q19. Calculate the skin depth and wave velocity in copper (sigma = 5.8 x 10^7 S/m, mu_r = 1) at: (a) power frequency 60 Hz, (b) radio frequency 1 MHz, (c) microwave frequency 10 GHz.

Q20. A Ground Penetrating Radar antenna on a concrete bridge deck detects a reflection from a reinforcing steel bar with a two-way travel time of 3.2 ns. If dry concrete has epsilon_r = 6.25 and mu_r = 1: (a) find wave velocity in concrete, (b) calculate the depth of the rebar, (c)

Q21. A plane wave in air with electric field amplitude 50 V/m impinges normally on a wet soil boundary (epsilon_r = 16, mu_r = 1, sigma = 0). Calculate: (a) reflection coefficient Gamma, (b) transmission coefficient tau, (c) reflected and transmitted electric field amplitudes, (d)

Q22. Explain the physical significance of complex permittivity and loss tangent. For seawater (epsilon_r = 80, sigma = 4.0 S/m), calculate the loss tangent and determine if it acts as a conductor or dielectric at: (a) 10 kHz, (b) 10 GHz.

Q23. Derive the magnetic vector potential A and electric scalar potential V wave equations in the Lorenz gauge from Maxwell's equations.

Q24. State and derive the electromagnetic boundary conditions for E, D, B, and H across an interface between two distinct dielectric media.

Q25. A 50 kW isotropic AM radio transmitter operates at 1.0 MHz. At a civil monitoring station 10 km away: (a) find the time-average Poynting vector magnitude, (b) calculate electric and magnetic field amplitudes, (c) determine the total EM energy stored within a 100 m x 100 m x 50 m airspace

Q26. Explain radiation pressure and electromagnetic momentum. Derive the expression for radiation pressure on a flat perfectly reflecting surface at normal incidence.

Q27. A 2.0 GHz microwave link passes through rain. If the rain-induced conductivity of air is sigma = 2.0 x 10^-4 S/m and epsilon_r = 1.05: (a) calculate the attenuation constant alpha in Np/m and dB/km, (b) calculate the phase velocity, (c) determine the percentage loss of power over a 15 km

Q28. Explain the physical principles and mathematical formulation of electromagnetic shielding effectiveness (SE) in civil data centers and shielded control rooms.

Q29. A lightning discharge delivers a current surge I(t) = 50*exp(-t / 10^-5) kA into a steel reinforced concrete foundation. Discuss the generation of transient magnetic and electric fields, skin effect in rebar, and induced step voltages.

Q30. Summarize the comprehensive impact of Maxwell's electromagnetic equations on modern civil engineering, geodesy, and structural health monitoring.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[B]	**Q41:**	[B]	**Q61:**	[B]	**Q81:**	[B]
Q2:	[D]	**Q22:**	[D]	**Q42:**	[D]	**Q62:**	[D]	**Q82:**	[D]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[C]	**Q24:**	[C]	**Q44:**	[C]	**Q64:**	[C]	**Q84:**	[C]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[C]	**Q26:**	[C]	**Q46:**	[C]	**Q66:**	[C]	**Q86:**	[C]
Q7:	[B]	**Q27:**	[B]	**Q47:**	[B]	**Q67:**	[B]	**Q87:**	[B]
Q8:	[D]	**Q28:**	[D]	**Q48:**	[D]	**Q68:**	[D]	**Q88:**	[D]
Q9:	[C]	**Q29:**	[C]	**Q49:**	[C]	**Q69:**	[C]	**Q89:**	[C]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[D]	**Q31:**	[D]	**Q51:**	[D]	**Q71:**	[D]	**Q91:**	[D]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[B]	**Q92:**	[B]
Q13:	[B]	**Q33:**	[B]	**Q53:**	[B]	**Q73:**	[B]	**Q93:**	[B]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[C]	**Q35:**	[C]	**Q55:**	[C]	**Q75:**	[C]	**Q95:**	[C]
Q16:	[D]	**Q36:**	[D]	**Q56:**	[D]	**Q76:**	[D]	**Q96:**	[D]
Q17:	[D]	**Q37:**	[D]	**Q57:**	[D]	**Q77:**	[D]	**Q97:**	[D]
Q18:	[C]	**Q38:**	[C]	**Q58:**	[C]	**Q78:**	[C]	**Q98:**	[C]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[B]	**Q80:**	[B]	**Q100:**	[B]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Summing over all volume elements in V, the internal boundary fluxes between adjoining interior cells cancel identically in pairs because outward normals on adjoining shared faces point in opposite directions.
- Only the flux across the external bounding surface S survives:
Closed_Integral_S (F dot dS) = Integral_V (Del dot F) dV. (Q.E.D.)

3. Application to Maxwell's First Equation:
- Gauss's Law in integral form: Closed_Integral_S (D dot dS) = Q_enclosed = Integral_V (rho_v) dV.
- Applying divergence theorem: Integral_V (Del dot D) dV = Integral_V (rho_v) dV.
- Since this holds for an arbitrary volume V: Del dot D = rho_v (Differential Form).

>>Solution Question 4: Derive the Equation of Continuity from the fundamental principles of charge conservation and Faraday's law in integral and differential forms. Explain its significance in time-varying

must flow outward across the surface S as an electric current.

2. Mathematical Formulation:
- Total enclosed charge: Q(t) = Integral_V rho(x,y,z,t) dV, where rho is volume charge density (C/m^3).
- Net outward current leaving through bounding surface S:
I_out = Closed_Integral_S (J dot dS), where J is current density vector (A/m^2).
- By charge conservation: I_out = - dQ / dt.
- Substituting expressions:
Closed_Integral_S (J dot dS) = - d/dt Integral_V rho dV = - Integral_V (d(rho)/dt) dV.

3. Application of Gauss's Divergence Theorem:
- Transform the surface integral into a volume integral:
Integral_V (Del dot J) dV = - Integral_V (d(rho)/dt) dV.
- Rearranging: Integral_V [Del dot J + d(rho)/dt] dV = 0.
- Since the volume V is completely arbitrary, the integrand must vanish identically everywhere:
Del dot J = - d(rho) / dt (Differential Equation of Continuity).

4. Physical Significance:
- In steady-state DC systems: d(rho)/dt = 0, which yields Del dot J = 0 (Kirchhoff's Current Law at a point).
- In AC and transient electrodynamic systems: d(rho)/dt != 0, proving that diverging conduction current implies a local depletion of charge.

>>Solution to Question 5: Explain the inconsistency of Ampere's Circuital Law for time-dependent fields. How did Maxwell resolve this dilemma by introducing the

- Modified Ampere-Maxwell Law: Del x H = J + dD / dt.

>>Solution Question Bank: Derive the Maxwell's equations in both differential and integral forms mathematically and explain the physical significance of each.

- Differential Form: Del x E = - dB / dt
- Integral Form: Closed_Integral_C (E dot dl) = - d/dt Integral_S (B dot dS) = - d(Phi_B) / dt
- Physical Basis: Faraday's and Lenz's laws of electromagnetic induction. A time-varying magnetic field produces an induced non-conservative, circulating electric field.

4. Maxwell's Equation IV (Ampere-Maxwell Circuital Law):
- Differential Form: Del x H = J_free + dD / dt
- Integral Form: Closed_Integral_C (H dot dl) = Integral_S (J_free + dD/dt) dot dS = I_conduction + I_displacement
- Physical Basis: Oersted-Ampere discovery extended by Maxwell. Magnetic fields are produced both by physical moving charges (conduction current) and by dynamically varying electric fields (displacement current).

>>**Solution Statement Question 8:** Derive the electromagnetic wave equation for electric field E in a charge-free, dielectric, and isotropic medium and state the speed of th

- Substitute Faraday's Law for Del x E:
Del x (Del x H) = epsilon * d/dt [- mu * (dH / dt)] = - mu * epsilon * (d^2H / dt^2).
- Apply vector identity to the left-hand side:
Del x (Del x H) = Del (Del dot H) - Del^2(H).
- By Gauss's Law for magnetism, Del dot B = 0 => Del dot H = 0. Therefore, Del (Del dot H) = 0.
- Substituting back gives:
- Del^2(H) = - mu * epsilon * (d^2H / dt^2) => Del^2(H) = mu * epsilon * (d^2H / dt^2).

3. Comparison of Phase Velocities:
- The phase velocity for H is v_H = 1 / sqrt(mu * epsilon).
- The phase velocity for E is v_E = 1 / sqrt(mu * epsilon).
- Both electric and magnetic wave components propagate in the medium with precisely the identical phase velocity v.
- In a non-magnetic medium where mu_r = 1: v = c / sqrt(epsilon_r) = c / n, where n = sqrt(epsilon_r) is the optical refractive index.

E(z,t) = E_0 * cos(omega*t - k*z) i.
- From transverse wave properties: E_y = 0, E_z = 0.

2. Application of Maxwell's Third Equation (Del x E = - mu_0 * dH/dt):
- Evaluate curl of E:
Del x E = i*(dE_z/dy - dE_y/dz) + j*(dE_x/dz - dE_z/dx) + k*(dE_y/dx - dE_x/dy).
Del x E = j * (dE_x / dz).
- From E_x(z,t): dE_x / dz = k * E_0 * sin(omega*t - k*z).
- Equating components with - mu_0 * (dH/dt):
- mu_0 * (dH_x/dt) = 0 => H_x = 0.
- mu_0 * (dH_y/dt) = dE_x/dz = k * E_0 * sin(omega*t - k*z).
- mu_0 * (dH_z/dt) = 0 => H_z = 0.
- Integrating H_y with respect to t:
H_y(z,t) = - (k * E_0 / mu_0) * Integral sin(omega*t - k*z) dt = (k * E_0 / (mu_0 * omega)) *
cos(omega*t - k*z).

3. Evaluating the Ratio and Intrinsic Impedance:
- Note that wave number k = omega / c = omega * sqrt(mu_0 * epsilon_0).
- Amplitude of H_y: H_0 = (k * E_0) / (mu_0 * omega) = [omega*sqrt(mu_0*epsilon_0) * E_0] /
(mu_0*omega) = sqrt(epsilon_0 / mu_0) * E_0.
- Therefore, the ratio of electric to magnetic field amplitude is defined as the intrinsic wave
impedance eta_0:
eta_0 = E_0 / H_0 = sqrt(mu_0 / epsilon_0).
- Numerical Calculation in Free Space:
mu_0 = 4*pi * 10^-7 H/m, epsilon_0 = 8.8541878 x 10^-12 F/m.
eta_0 = sqrt((4*pi * 10^-7) / (8.8541878 x 10^-12)) = 376.73 Ohms (~ 120*pi Ohms ~ 377 Ohms).

>>Solution to Question 12: and derive Poynting's Theorem from Maxwell's equations. Provide the physical interpretation of each term in the resulting equatio

- Applying Gauss's Divergence Theorem to the left-hand side:
Closed_Integral_S (S dot dS) = - d/dt Integral_V u_EM dV - Integral_V (J dot E) dV.

3. Physical Interpretation of Terms:
- Closed_Integral_S (S dot dS): Rate of electromagnetic energy leaving volume V through boundary surface S.
- d/dt Integral_V u_EM dV: Time rate of decrease of stored electric and magnetic energy within volume V.
- Integral_V (J dot E) dV: Rate of energy converted into Joule heat (ohmic dissipation) or mechanical work done on charges.

- Take curl of Eq (3): Del x (Del x E) = - mu * d/dt (Del x H).
- Substitute Eq (4):
Del(Del dot E) - Del^2(E) = - mu * d/dt [sigma * E + epsilon * (dE/dt)].
- Since Del dot E = 0:
Del^2(E) = mu * sigma * (dE / dt) + mu * epsilon * (d^2E / dt^2) [Telegrapher's Equation for E].

3. Time-Harmonic Fields (d/dt -> j*omega):
- Del^2(E) = [j*omega*mu*sigma - omega^2*mu*epsilon] E = gamma^2 * E,
where gamma is the complex propagation constant: gamma = alpha + j*beta = sqrt(j*omega*mu*(sigma + j*omega*epsilon)).

4. Expressions for alpha and beta:
- Squaring both sides: (alpha + j*beta)^2 = alpha^2 - beta^2 + 2*j*alpha*beta = - omega^2*mu*epsilon + j*omega*mu*sigma.
- Equating real and imaginary parts:
(i) beta^2 - alpha^2 = omega^2 * mu * epsilon
(ii) 2*alpha*beta = omega * mu * sigma
- Solving the coupled equations yields:
alpha = omega * sqrt((mu*epsilon / 2) * [sqrt(1 + (sigma / (omega*epsilon))^2) - 1])
beta = omega * sqrt((mu*epsilon / 2) * [sqrt(1 + (sigma / (omega*epsilon))^2) + 1]).

>>Solution to Question 15: Define skin depth (depth of penetration). Derive its formula for a good conductor and explain its engineering consequences in structura

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- Electric field between plates: E(t) = V(t) / d.
- J_d(t) = epsilon_0 * (dE/dt) = (epsilon_0 / d) * (dV/dt) = (C / A) * (dV/dt) = i_c(t) / A.
- Peak J_d: J_d0 = 2.224 x 10^-5 / 0.070686 = 3.146 x 10^-4 A/m^2 = 314.6 microamperes/m^2.
- Expression: J_d(t) = 314.6 * cos(100*pi*t) microamperes/m^2.

5. (d) Induced Magnetic Field B at r = 0.10 m:
- By Ampere-Maxwell law for a circular path of radius r <= R:
Closed_Integral (B dot dl) = mu_0 * i_d,enclosed => B * (2*pi*r) = mu_0 * J_d * (pi * r^2).
- B(r,t) = (mu_0 * J_d(t) * r) / 2.
- Peak B: B_0 = (4*pi x 10^-7 * 3.146 x 10^-4 * 0.10) / 2 = 1.977 x 10^-11 Tesla = 19.77 pT.
- Result: B(r,t) = 19.77 * cos(100*pi*t) picotesla.

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3. (b) At Radio Frequency f = 1 MHz (10^6 Hz):
- delta = 0.0661 / sqrt(10^6) = 0.0661 / 1000 = 6.61 x 10^-5 m = 66.1 micrometers.
- v = 2 * pi * 10^6 * 6.61 x 10^-5 = 4.15 x 10^2 m/s = 415 m/s.

4. (c) At Microwave Frequency f = 10 GHz (10^10 Hz):
- delta = 0.0661 / sqrt(10^10) = 0.0661 / 10^5 = 6.61 x 10^-7 m = 0.661 micrometers.
- v = 2 * pi * 10^10 * 6.61 x 10^-7 = 4.15 x 10^4 m/s = 41.5 km/s.

5. Engineering Observation:
- Skin depth decreases by four orders of magnitude from 8.5 mm down to 0.66 microns.
High-frequency waveguides and RF cables need only thin surface silver/gold plating.

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3. (b) Transmission Coefficient tau:
- tau = (2 * eta_2) / (eta_2 + eta_1) = (2 * 94.25) / 471.25 = 188.5 / 471.25 = +0.40.
- Check: 1 + Gamma = 1 + (-0.60) = 0.40 = tau (consistent).

4. (c) Reflected and Transmitted Field Amplitudes:
- E_r0 = |Gamma| * E_i0 = 0.60 * 50 = 30 V/m (with 180 degree phase reversal).
- E_t0 = tau * E_i0 = 0.40 * 50 = 20 V/m.

5. (d) Power Reflection (R) and Transmission (T):
- Reflectance R = |Gamma|^2 = (-0.60)^2 = 0.36 = 36%.
- Transmittance T = (eta_1 / eta_2) * tau^2 = (377 / 94.25) * (0.40)^2 = 4 * 0.16 = 0.64 = 64%.
- Energy conservation check: R + T = 0.36 + 0.64 = 1.00 (100%).

- An irrotational field can be written as the gradient of a scalar potential V:
E + dA/dt = - Del V => E = - Del V - dA/dt.

3. Substituting into Maxwell's First Equation:
- Del dot E = rho / epsilon => Del dot (- Del V - dA/dt) = rho / epsilon.
- - Del^2(V) - d/dt (Del dot A) = rho / epsilon => Del^2(V) + d/dt (Del dot A) = - rho / epsilon.

4. Substituting into Maxwell's Fourth Equation:
- Del x B = mu*J + mu*epsilon * (dE/dt).
- Del x (Del x A) = mu*J + mu*epsilon * d/dt (- Del V - dA/dt).
- Del(Del dot A) - Del^2(A) = mu*J - mu*epsilon*Del(dV/dt) - mu*epsilon*(d^2A/dt^2).
- Rearranging: Del^2(A) - mu*epsilon*(d^2A/dt^2) - Del [Del dot A + mu*epsilon*(dV/dt)] = - mu*J.

5. The Lorenz Gauge Condition:
- Set: Del dot A + mu*epsilon*(dV/dt) = 0 (Lorenz Gauge).
- The equations decouple completely into symmetric inhomogeneous wave equations:
Del^2(A) - mu*epsilon*(d^2A/dt^2) = - mu*J
Del^2(V) - mu*epsilon*(d^2V/dt^2) = - rho / epsilon.

[Detailed Step-by-Step Answer]:
1. (a) Time-Average Poynting Vector Magnitude S_avg:
- Transmitter radiates power P = 50 kW = 50,000 W uniformly over a sphere of radius R = 10,000 m.
- Sphere surface area: A_sphere = 4 * pi * R^2 = 4 * pi * (10,000)^2 = 1.2566 x 10^9 m^2.
- S_avg = P / A_sphere = 50,000 / (1.2566 x 10^9) = 3.979 x 10^-5 W/m^2 = 39.79 microwatts/m^2.

2. (b) Electric and Magnetic Field Amplitudes:
- S_avg = 0.5 * (E_0^2 / eta_0) => E_0 = sqrt(2 * eta_0 * S_avg).
- E_0 = sqrt(2 * 377 * 3.979 x 10^-5) = sqrt(0.0300) = 0.1732 V/m = 173.2 mV/m.
- H_0 = E_0 / eta_0 = 0.1732 / 377 = 4.594 x 10^-4 A/m = 459.4 microamperes/m.
- Peak B field: B_0 = mu_0 * H_0 = 4*pi x 10^-7 * 4.594 x 10^-4 = 5.77 x 10^-10 T = 0.577 nT.

3. (c) Total Stored EM Energy in Volume V:
- Volume V = 100 m * 100 m * 50 m = 500,000 m^3.
- Average energy density: <u_total> = S_avg / c = (3.979 x 10^-5 W/m^2) / (3.0 x 10^8 m/s) = 1.326 x 10^-13 J/m^3.
- Total stored energy: U = <u_total> * V = 1.326 x 10^-13 * 500,000 = 6.63 x 10^-8 Joules = 66.3 nanojoules.

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1. Given Parameters:
- Frequency f = 2.0 GHz = 2.0 x 10^9 Hz; omega = 2*pi*f = 1.2566 x 10^10 rad/s.
- sigma = 2.0 x 10^-4 S/m; epsilon_r = 1.05; mu_r = 1.0.
- epsilon = 1.05 * 8.854 x 10^-12 = 9.297 x 10^-12 F/m.
- Loss tangent: tan(delta) = sigma / (omega*epsilon) = 2.0 x 10^-4 / (1.2566 x 10^10 * 9.297 x 10^-12) = 2.0 x 10^-4 / 0.1168 = 1.712 x 10^-3 << 0.1 (low-loss dielectric).

2. (a) Attenuation Constant alpha:
- For low-loss dielectric: alpha ~= 0.5 * sigma * sqrt(mu / epsilon) = 0.5 * sigma * (eta_0 / sqrt(epsilon_r)).
- alpha = 0.5 * (2.0 x 10^-4) * (377 / sqrt(1.05)) = 1.0 x 10^-4 * (377 / 1.0247) = 3.679 x 10^-2 Np/m.
- In dB/km: Attenuation (dB/km) = 8.686 * alpha * 1000 = 8.686 * 0.03679 * 1000 = 319.6 dB/km.

3. (b) Phase Velocity v:
- v ~= c / sqrt(epsilon_r) = (3.0 x 10^8) / sqrt(1.05) = (3.0 x 10^8) / 1.0247 = 2.928 x 10^8 m/s.

4. (c) Power Loss over d = 15 km:
- Total attenuation in dB: A_total = 319.6 dB/km * 15 km = 4,794 dB.
- Total power transmitted is practically 0% (complete signal extinction due to severe heavy rain absorption at 2 GHz).

>> Solution to Question 28: Explain the physical principles and mathematical formulation of electromagnetic shielding effectiveness (SE) in civil data centers and sh

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3. Conclusion:
- Maxwell's electromagnetic theory is not merely pure physics; it is an indispensable engineering foundation for the design, construction, diagnostic maintenance, and lifecycle safety of modern civil infrastructure.
